

Where Should America's Next Sustainable Data Center Be Built?

The Challenge

AI is reshaping the world—and creating unprecedented demand for data center infrastructure.

As AI workloads accelerate, the world needs to build more digital infrastructure. But where and how we build that infrastructure matters.

Data centers consume significant amounts of electricity, water, land, materials, and other resources. A location that looks attractive today may create decades of environmental, operational, and economic challenges.

Where should we build the next generation of AI infrastructure—and what makes a location truly future-ready?

Your mission is to create a decision engine and use data, technology, and systems thinking to identify the optimal geographic location for a next-generation sustainable AI data center, balancing environmental impact, infrastructure availability, resilience, and long-term economic viability.

Inspired by the principles of open, efficient, scalable infrastructure, teams will explore how smarter location decisions can help accelerate a more sustainable digital future.

The Problem

The location of a data center creates a complex interaction between energy, water, climate, infrastructure, communities, and the broader industrial ecosystem.

Teams should consider factors such as:

- **Energy & Carbon** — Grid carbon intensity, renewable energy availability, clean-energy penetration, and future energy capacity
- **Heat Recovery & Reuse** — Opportunities to capture and reuse waste heat from the data center for district heating, greenhouses, industrial processes, agriculture, aquaculture, or other beneficial applications
- **Water** — Water availability, scarcity, cooling requirements, and long-term water risk. Consider water used in the generation of grid power as well.
- **Climate Resilience** — Exposure to extreme heat, flooding, hurricanes, drought, wildfires, and other climate risks
- **Grid & Infrastructure** — Grid reliability, transmission capacity, energy costs, connectivity, and infrastructure readiness
- **Land & Materials** — Land availability, potential land impacts including proximity to sensitive areas and ecosystems, and opportunities to reduce the embodied carbon of construction

- Community & Economic Impact — Workforce availability, economic development opportunities, community impact, and proximity to users and other infrastructure

Your Objective

Find the world's most future-ready location for sustainable AI infrastructure.

Develop a data-driven decision framework that identifies where a next-generation AI data center should be located and demonstrates why that location outperforms the alternatives.

Your solution should balance environmental sustainability, infrastructure performance, resource efficiency, resilience, and economic viability—not optimize for a single metric.

Think beyond the building.

Think about the ecosystem.

OCP Relevance

Build the infrastructure of the future—before the future arrives.

OCP has demonstrated that data center infrastructure can be approached differently through open collaboration, innovation, efficiency, and scalable design.

Can we apply the same systems-thinking approach to where infrastructure is built?

Use open data, innovative technologies, and creative thinking to rethink the geography of AI infrastructure.

Don't just find a location. Design the decision framework that could change how the world builds data centers.

Deliverables

Teams should provide:

1. Recommended Geographic Location: city, region, or country
2. Decision Framework: Methodology and weighting of criteria
3. Data Analysis: Supporting datasets, assumptions, and insights
4. Sustainability Impact Assessment: Estimated carbon, water, and resource benefits
5. Risk Assessment: Key environmental and operational risks
6. Implementation Vision: How the data center could operate sustainably over the next 20-30 years

Major Considerations

Your solution should evaluate and balance factors such as:

Energy & Carbon

- Availability of renewable energy [NSRDB](#)
- Availability of wind energy [WRDB](#)
- Carbon intensity of the electrical grid [Emissions & Generation Resource Integrated Database \(eGRID\) | US EPA](#)
- Future grid decarbonization potential [Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection | Energy Markets & Planning](#)
- Grid reliability and resilience [Annual Electric Power Industry Report, Form EIA-861 detailed data files - U.S. Energy Information Administration \(EIA\)](#)
- Fuel Cells: BloomEnergy

Water Stewardship

- Water availability and stress levels [Aqueduct Data | World Resources Institute](#)
- Cooling requirements [U.S. Climate Normals | National Centers for Environmental Information \(NCEI\)](#)
- Alternative cooling technologies [Clean Watersheds Needs Survey | US EPA](#)
- Long-term water security [Clean Watersheds Needs Survey | US EPA](#)
- Fuel Cells: BloomEnergy - carbon capture and byproduct

Construction & Embodied Carbon

- Availability of low-carbon steel [Global Iron and Steel Tracker](#)
- Availability of low-carbon concrete alternatives [Global Cement and Concrete Tracker](#)
- Construction material sourcing impacts [Freight Analysis Framework | Bureau of Transportation Statistics](#)
- Embodied carbon footprint [Tools - Building Transparency](#)

Land & Infrastructure

- Suitable land availability [Annual NLCD Data Access | U.S. Geological Survey](#)
- Existing infrastructure access [US Energy Atlas](#)
- Fiber network connectivity [FCC Broadband Map](#)
- Transportation and logistics considerations [National Transportation Atlas Database](#)

Climate Resilience

- Flood risk [Flood Data Viewers and Geospatial Data | FEMA.gov](#)
- Wildfire risk [Wildfire Risk to Communities](#)
- Hurricane and storm exposure [International Best Track Archive for Climate Stewardship \(IBTrACS\) | National Centers for Environmental Information \(NCEI\)](#)
- Drought susceptibility [Drought Monitor](#)
- Extreme temperature impacts [Resilience Analysis and Planning Tool \(RAPT\) | FEMA.gov](#)
- Long-term climate change projections [NASA Earth Exchange Global Daily Downscaled Projections \(NEX-GDDP-CMIP6\) – NCCS](#)
- Biome: Environment and biodiversity
- IMasons Climate Accord

